Rotaract Club of Xiamen University Malaysia

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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	overinfinity
Project / Solution Name	AURA-Workforce
Name (Leader)	Ahmad Adam Hakimi bin Naeman
Email & Phone (Leader)	0damkim254@gmail.com / +60 18-344 5135
University Affiliation(s)	International Islamic University Malaysia (IIUM)
Live Prototype URL	https://aura-workforce.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/adamnaemman/sdc-hackathon

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

AURA-Workforce is a two-service deployment on Replit: a React single-page app in the browser and an Express API mounted at /api. The API serves three things. First, a curated catalogue of five future-ready roles and their learning steps, each step pointing at a free, verified provider page. Second, a live trend layer: it fetches recent public job postings, normalises the text, counts how often each role's required skills appear, and returns a demand share stamped with source, fetch time and sample size; results are cached for six hours. Third, two AI endpoints on DeepSeek: one expands the person's role, industry and experience into concrete skills, and one receives the whole profile plus the catalogue and returns a grounded recommendation with an ordered pathway. Every role ID and module ID returned by the model is validated against the catalogue before it reaches the browser. No personal data is stored anywhere; the profile lives in browser memory only.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
React 19.1	User interface	MIT
Vite 7	Build tool and dev server	MIT
Tailwind CSS 4	Styling system	MIT
Radix UI primitives	Accessible UI components	MIT
TanStack Query 5	Client data fetching and cache	MIT
Express 5	HTTP API server	MIT
Zod 3	Request and response validation	MIT

Name & Version	Purpose in Project	License
Orval 8	OpenAPI to typed client generation	MIT
TypeScript 5.9	Language and type checking	Apache-2.0
pnpm 10	Workspace package manager	MIT
Lucide	Icon set	ISC

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
DeepSeek API (api.deepseek.com)	Language model for profile extraction and the AI planner	Paid (team-funded)
Arbeitnow Job Board API	Live source of public job postings for the trend layer	Free, no key

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Core (deployment + secrets)	Hosts the web app and the API; stores API keys as secrets	Sponsor-provided, Singapore region
GitHub	Source control and public repository for judges	Free
RunPod (reserved)	GPU credit reserved for Phase 2 model hosting; not used in this prototype	Sponsor-provided, USD 20/team

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
deepseek-chat	DeepSeek	Pre-trained (no fine-tuning)	Commercial API terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
U.S. BLS Occupational Outlook Handbook (2025-2035 projections)	https://www.bls.gov/ooh/	Public	Public domain (US Government)	N
WEF Future of Jobs Report 2025	https://www.weforum.org/publications/the-future-of-jobs-report-2025/	Public	Quoted with attribution	N
Arbeitnow Job Board API (live postings)	https://www.arbeitnow.com/api/job-board-api	Public	Public API, free, no key	N
AURA role and learning-step catalogue	https://github.com/adamnaemman/sdc-hackathon	Synthetic / curated by the team	Team-owned, sources cited in app	N
Hays Asia Salary Guide 2025 (two figures, quoted)	Cited in PRD section 2.1	Public	Quoted with attribution	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
O*NET 30.1 database (technology skills reference)	https://www.onetcenter.org/database.html	Public	CC BY 4.0	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
DeepSeek (chat + coding agent)	Code generation, refactoring, debugging, documentation	Major
Replit Agent	Answered Git workflow questions only; no code committed from it	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every AI-generated change was reviewed as a diff before running, then verified by a workspace-wide TypeScript typecheck and a successful build in Replit. Endpoints were tested with scripted requests, including a prompt-injection attempt that the model correctly ignored. Generated copy was checked against the catalogue for invented roles, courses and statistics, and two data bugs were found and fixed this way. The team can explain every file, and the scoring engine remains deterministic.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core, one month per team member	Hosting, deployment and secret storage for the prototype
RunPod	USD 20 GPU credit per team	Not used in the preliminary prototype; reserved for Phase 2 model hosting

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit Core is essential: the prototype is built, run and deployed there, and it hosts both the API and the web app. The RunPod credit is not used yet; the AI client is OpenAI-compatible, so moving the model to a self-hosted GPU on RunPod in Phase 2 is a configuration change rather than a rewrite. The DeepSeek API is paid by the team, not sponsor-provided.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The application collects no account data and stores nothing on the server. A profile exists only in browser memory and is discarded when the tab closes. Two things leave the browser, both triggered by a button: a one-line summary of role, industry and years of experience (sent to our API, forwarded to DeepSeek, neither logged nor stored), and the structured profile sent with the AI plan request. Neither contains a name or contact detail. There is no CV upload and no database. Server logs record method, URL, status and response time only.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Hardest risk: the model inventing a job, course or statistic. Mitigation: it may only choose from a five-role, twenty-step catalogue, every returned ID is validated server-side, and its fit score is bounded and displayed next to the deterministic estimate. Second risk: the live job feed is a global, tech-oriented board, so it measures the direction of skill demand rather than Malaysian vacancy counts; the interface states this and shows the sample size. Third: five roles cannot serve every occupation, so the plan is required to say when no role is a strong match. Fourth: a single AI provider; the client is OpenAI-compatible, so a fallback endpoint is a configuration change.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Ahmad Adam Hakimi bin Naeman (Team Leader, overinfinity)

23 October 2026